Chapter 9, Problem 74P

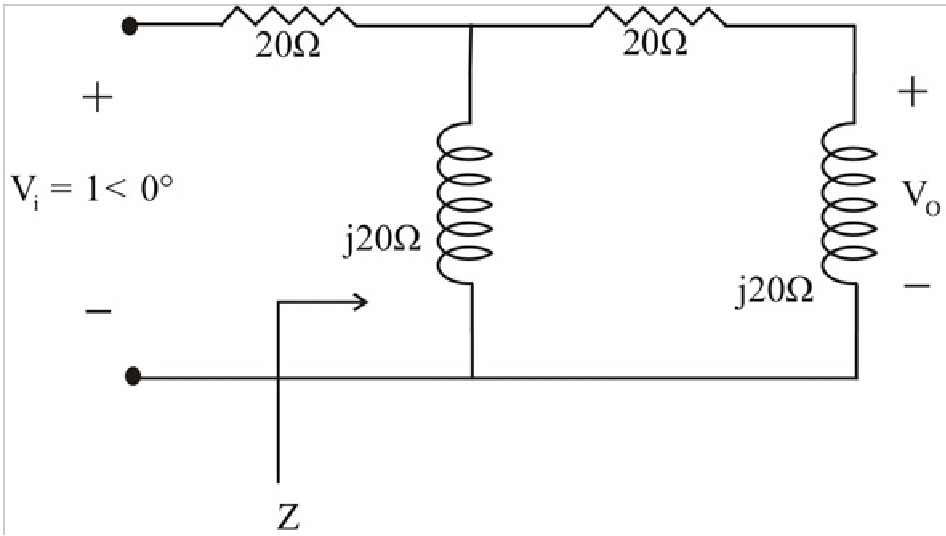
Problem

Design an *RL* circuit to provide a 90° leading phase shift.

Step-by-step solution

Step 1 of 2

One such RL circuit is shown below.



Step 2 of 2

We now want to show that this circuit will produce a $90°$ phase shift.

$$Z = j20 \parallel (20 + j20)$$

$$= \frac{(j20)(20+j20)}{20+j40} = \frac{-20+j20}{1+j20}$$

$$= 4(1+j3)$$

$$V = \frac{Z}{Z+20}V_i = \frac{4+j2}{24+j12}(1\angle 0°)$$

$$= \frac{1+j3}{6+j3} = \frac{1}{3}(1+j)$$

$$V_0 = \frac{j20}{20+j20} - V = \left(\frac{j}{1+j}\right)\left(\frac{1}{3}(1+j)\right)$$

$$= \frac{1}{3} = 0.3333\angle 90° \ .$$

This shown that the output leads the input by $90°$.